OpenUBEM European Locations — MVP Implementation Specification

Architectural Integration of European Standards, Building Typologies, Procedural Floor Layouts, and BEM Simulation Coupling with GSSCanada

- **Document Version**: `1.0.0-PROD`
- **Target Subsystem**: `openubem.data`, `openubem.geometry`, `openubem.idf`, `openubem.semantic`, `openubem.simulation`
- **Location in Repo**: `docs/docs_ACTIVE/europeanLocations/MVP_european_locations.md`
- **Sister Walkthrough**: `docs/docs_ACTIVE/europeanLocations/WALKTHROUGH_european_locations.md`
- **GSSCanada Reference**: `C:\Users\o_iseri\Desktop\GSSCanada\GSSCanada-main\4J_docs_occ\Step8_docs\IMP_step8\4thJ_08_bemSimulation_IMP.md`
- **Scientific Provenance**: *Iseri et al. (2025), Energy and Buildings 337, 115620*; `IMP_step8/outputs/floor_layout_generation_report.md`, `IMP_step8/outputs/kbem_ankara_report.md`, `IMP_step8/outputs/simulation_results_analysis_report.md`
- **Core OpenUBEM Docs**: `OpenUBEM_fundamentals.md`, `OpenUBEM_inputs_reference.md`, `OpenUBEM_imputation_methods.md`, `simulated_vs_reconstructed_methodology.md`, `OpenUBEM_debug_References.md`
- **Reusable Figure/Table Assets**: `content/`

> **Document role.** This MVP is the principal implementation specification. It owns scientific decisions, scope, interfaces, data contracts, algorithms, acceptance criteria, and the definition of done. The sister walkthrough owns ordered tasks, runnable commands, stop conditions, and the append-only progress log; it must link back here instead of creating a second scientific contract.

> **Implementation-status notice (v1.1 review, 2026-08-22).** The original v1.0 text below is retained in full for provenance. It describes the intended target architecture, not the current repository state. The authoritative, code-audited delta is in **Section 9**. Until the listed implementation and validation work is complete, examples and `[x] PASS`-style claims in the original text must be read as design intent, not evidence of an executed Step 8 campaign.
>
> **Speed/pre-occupant extension.** Section 10 adds the Speed HPC execution profiles and the required Q0–Q3 physics qualification sequence before any `f>0` occupant schedule is authorized.

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1. Executive Summary & Strategic Rationale

1.1 Why OpenUBEM Must Be Built and Adapted *Before* GSSCanada BEM Simulations

A foundational principle governs this integration: **OpenUBEM is the physical foundation and computational simulation engine; GSSCanada is the demographic occupancy schedule provider.**

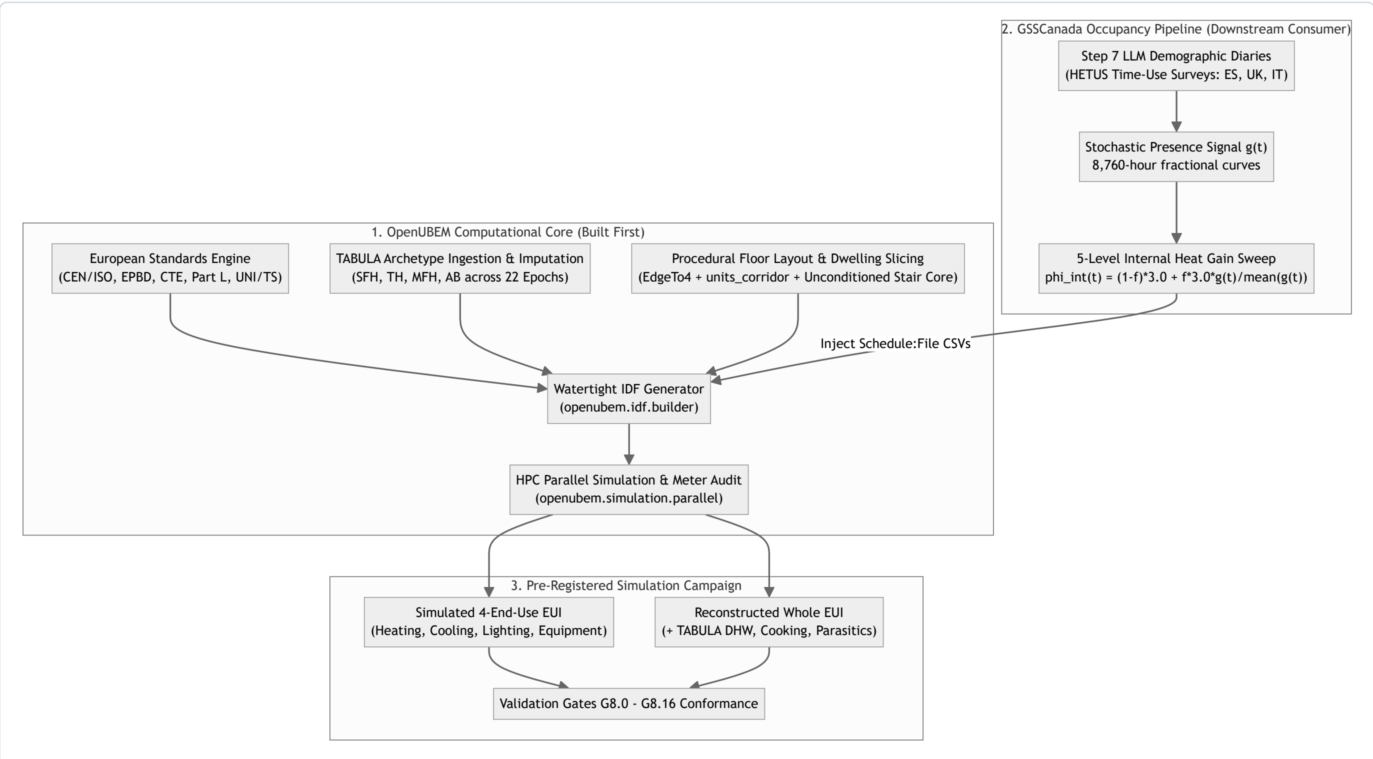

Figure 1. OpenUBEM provides the European physical model and GSSCanada provides held-out occupant-presence inputs; both meet at the versioned campaign-cell interface before independent validation. Reusable source: `content/figure_1_1_integration_pipeline.mmd`.

Attempting to run GSSCanada European BEM simulations without first establishing European physics, building stock definitions, and procedural layout geometry within OpenUBEM results in complete methodological collapse for four fundamental reasons:

1. **Thermodynamic Distortion of North American Defaults**: OpenUBEM's baseline configuration references North American commercial and multi-family standards (ASHRAE Standard 90.1, DOE Prototypes, IECC). US construction defaults feature lightweight wood-stud and steel-frame assemblies with negligible thermal mass, forced-air packaged DX

cooling/heating, and commercial continuous ventilation rates (0.8--1.5 ACH). Injecting European demographic occupancy profiles into lightweight US structures causes severe, non-physical indoor temperature spikes and distorts heating demand by $> 35\%$, because the structural thermal capacitance (c_m) that buffers real European masonry buildings is completely absent.

2. **Spatial Topology & Inter-Dwelling Heat Transfer**: European residential stocks (especially Multi-Family Houses `MFH` and Apartment Blocks `AB`) consist of compartmentalized dwelling units arranged around central unconditioned staircase cores. Empirical research across **6,458 dwelling units** (*Iseri et al., 2025*) proves that coarse building-level modeling **suppresses $> 75.5\%$ of inter-dwelling energy variance** (standard deviation drops from 63.61 to 15.54 $\mathrm{kWh/m^2a}$) and underestimates peak thermal vulnerability by a factor of $3.2\times$. Modeling a whole floor as a single lumped zone erases party-wall conduction between adjacent flats (which accounts for 15%--40% of heat loss when units have differing occupancy or setpoints) and fails to capture the thermal buffering of unconditioned circulation spaces ($b_u = 0.50$--0.80).
3. **Strict Separation of Concerns (Engine vs. Domain Scenario)**: OpenUBEM is designed as a reusable, reproducible, pip-installable Python package (`openubem`). GSSCanada Step 8 is a scientific scenario driver that tests demographic hypotheses on European housing. Building the geometry generators, TABULA construction translators, and schedule ingestion hooks inside `openubem` ensures that OpenUBEM remains a general-purpose Urban Building Energy Modeling framework, while GSSCanada simply orchestrates campaign sweeps via clean API calls.
4. **Pre-Registered Validation Gate Compliance**: The 4J HETUS Step 8 pre-registered protocol mandates strict validation gates (Gate **G8.0** uninjected control at $f = 0.00$, Gate **G8.8** scenario differentiation, Gate **G8.13** `Interpolate to Timestep = No` on `Schedule:File` objects, and Gate **G8.10** meter energy balance). These gates require low-level structural assertions in OpenUBEM's IDF builder, EnergyPlus output dictionary parser, and manifest logger.

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2. European Building Standards & Energy Physics Framework

2.1 Pan-European Normative Stack (CEN / ISO / EPBD)

OpenUBEM's European engine parameterizes building physics in strict compliance with the European Committee for Standardization (CEN) and International Organization for Standardization (ISO) standards hierarchy:

Standard / directive	Scope and implementation role
EPBD (EU) 2024/1275	European energy-performance framework; the applicable national implementation remains authoritative.
EN ISO 52016-1:2017	Hourly heating/cooling needs and zone heat-balance framework.
EN 16798-1:2019	Indoor-environment input parameters and comfort categories.
EN 410 / EN 673	Glazing solar and thermal properties, including g_{gl} and U_w.
ISO 18523-2:2018	Residential usage schedules as a contextual reference; it does not replace held-out Step 7 diaries.
TABULA / EPISCOPE	Residential typology structure and national example-building data. The ES/GB/IT occupant campaign has 102 target records; France is included in the physical-model branch and deferred only for occupant schedules.

Table 1. Normative sources used to define or contextualize European residential model inputs. Machine-readable copy: [`content/table_2_1_normative_stack.csv`].

2.2 Country-Specific Statutory Codes & Archetype Crosswalk

Spain, England-limited `GB`, and Italy form the current **occupant-schedule** campaign. France is in scope now for residential stock preparation, neighbourhood generation, geometry/IDF construction, weather preparation, and baseline physical simulation. France-specific occupant schedules and the five-level occupant-effect sweep remain future work, so France does not yet change the frozen ES/GB/IT occupant counts or 510-run total.

Dimension	Spain (`ES`)	England-limited stock (`GB`)	Italy (`IT`)	France (`FR`) — physical now / occupants later
Building/neighbourhood pipeline	`IN_SCOPE`	`IN_SCOPE`	`IN_SCOPE`	`IN_SCOPE`
Baseline physical simulation	`IN_SCOPE`	`IN_SCOPE`	`IN_SCOPE`	`IN_SCOPE_AFTER_FR_REGISTRY_AUDIT`
Occupant-schedule sweep	`IN_SCOPE`	`IN_SCOPE`	`IN_SCOPE`	`FUTURE_SCOPE`
Primary new-build code	CTE DB-HE	Building Regulations Part L	DM 26/06/2015	RE2020
Regulatory calculation context	HULC	SAP	UNI/TS 11300	Th-BCE 2020
Existing-home energy record	CEE/EPC	EPC	APE	DPE
Campaign archetype count	24	36	42	`NOT_AUDITED`
Weather and diary alignment	Required and unresolved until manifest acceptance	Required and unresolved until manifest acceptance	Required and unresolved until manifest acceptance	`NOT_SELECTED`
Effect on occupant-campaign totals	Included in 102/510	Included in 102/510	Included in 102/510	No effect on 102/510 until French occupant inputs are approved

Table 2. Regulatory and campaign-scope crosswalk. France is included in the physical building/neighbourhood pipeline; only its occupant-schedule sweep is deferred. Regulatory values remain candidates until the France registry and parameter provenance pass acceptance. Sources: French RE2020 consolidated texts, Th-BCE 2020 engine, French DPE, and TABULA France country page. Machine-readable copy: [`content/table_2_2_country_crosswalk.csv`].

2.2.1 France Physical-Pipeline Gate (`FR-PHYS`) and Occupant Deferral (`FR-OCC-FUTURE`)

France proceeds now through acquisition, residential filtering, semantic mapping, procedural layouts, envelope/HVAC/weather preparation, saved-IDF audits, and baseline physical simulations. Before those baseline simulations are accepted, the France branch must reproduce an authoritative residential typology registry, define construction-period crosswalks, select baseline weather, validate national envelope/HVAC assumptions, and pass the same geometry, saved-IDF, meter, warning, and mutation gates. TABULA/EPISCOPE documents a French residential typology, and an EPISCOPE synthesis describes 26 French building families, but the exact production subset and count remain `NOT_AUDITED` for OpenUBEM.

The France occupant branch is separately labelled `FR-OCC-FUTURE`. It begins only when a France-compatible held-out diary/model source, leakage controls, fieldwork-aligned weather window, and schedule manifest are accepted. Until then, French baseline runs must use the controlled non-stochastic schedule path and must not be merged into the ES/GB/IT 510-case occupant-effect denominator.

2.2.2 European Ventilation Rate Equivalence (EN 16798-1 Table B.4)

EN 16798-1 Table B.4 specifies baseline residential fresh air rates of $0.23\text{--}0.35\ \mathrm{L/(s\cdot m^2)}$ (approximately $0.30\text{--}0.60\ \mathrm{ACH}$). The country-specific values above are TABULA boundary-condition implementations of this range.

2.2.3 Intermittent Heating Reduction Factors ($F_{\text{red,htr}}$)

European standards apply intermittent heating correction as transmission multipliers on UA, not as thermostat night-setback schedules (TABULA `Tab.BoundaryCond`; EN ISO 13790:2008 Section 13.2): - Single-Use Housing (`EU.SUH`): $F_{\text{red,htr1}} = 0.90$, $F_{\text{red,htr4}} = 0.80$ - Multi-Use Housing (`EU.MUH`): $F_{\text{red,htr1}} = 0.95$, $F_{\text{red,htr4}} = 0.85$

These are preserved as scalar transmission reductions in the campaign. **No scheduled thermostat night-setback is added**, to prevent confounding with the LLM-generated stochastic occupancy signal (Pre-Registered Ruling `D-S8-2`).

2.2.4 Italian Area-Dependent Internal Gain Formula (UNI/TS 11300-1 Table 1)

While the campaign uses the harmonized TABULA $3.0\ \mathrm{W/m^2}$ baseline, the Italian national standard defines a more nuanced area-dependent formula:

$$\Phi_{\text{int}} = 4.0\ \mathrm{W/m^2} \quad (A_{\text{floor}} \le 120\ \mathrm{m^2}); \qquad \Phi_{\text{int}} = 4.0 \cdot \left(\frac{120}{A_{\text{floor}}}\right)^{0.2}\ \mathrm{W/m^2} \quad (A_{\text{floor}} > 120\ \mathrm{m^2})$$

This is recorded as contextual information, not as the Step 8 campaign baseline.

2.2.5 Overheating Assessment Standards

European residential overheating is assessed using the adaptive comfort model (EN 16798-1 Section 6.2, Annex A) and cumulative degree-hours above threshold (IOD). The UK additionally requires compliance with **CIBSE TM59 (2017)**: - *Criterion A*: Living/bedrooms must not exceed $\Delta T \ge 1\ \mathrm{K}$ above the comfort limit for more than 3% of occupied hours. - *Criterion B*: Bedrooms must not exceed $26^\circ\mathrm{C}$ for more than $32\ \mathrm{hours}$ between 22:00–07:00.

These criteria require hourly zone-level operative temperatures, which are available from the EnergyPlus dwelling-level output.

2.3 Physical Parameter Ingestion & Translation

2.3.1 Parametric Opaque Assembly Sizing (`openubem.idf.opaque_assembly`)

Unlike North American models with fixed nominal lumber or steel studs, European envelope retrofits and epoch standards vary widely in continuous insulation thickness. OpenUBEM sizes insulation dynamically:

$$d_{\text{ins}} = \lambda_{\text{ins}} \cdot \left(\frac{1}{U_{\text{target}}} - R_{\text{si}} - R_{\text{se}} - \sum_j \frac{d_j}{\lambda_j}\right)$$

Where standard European material thermal properties are applied: - Outer clay brick masonry: $d = 0.20\ \mathrm{m}$, $\lambda = 0.79\ \mathrm{W/(m\cdot K)}$, $\rho = 1800\ \mathrm{kg/m^3}$, $c_p = 1000\ \mathrm{J/(kg\cdot K)}$ - Reinforced concrete panel: $d = 0.15\ \mathrm{m}$, $\lambda = 1.40\ \mathrm{W/(m\cdot K)}$, $\rho = 2300\ \mathrm{kg/m^3}$, $c_p = 1000\ \mathrm{J/(kg\cdot K)}$ - Expanded Polystyrene (EPS) insulation: $\lambda_{\text{ins}} = 0.038\ \mathrm{W/(m\cdot K)}$, $\rho = 25\ \mathrm{kg/m^3}$, $c_p = 1400\ \mathrm{J/(kg\cdot K)}$ - Interior gypsum plaster: $d = 0.015\ \mathrm{m}$, $\lambda = 0.25\ \mathrm{W/(m\cdot K)}$, $\rho = 900\ \mathrm{kg/m^3}$, $c_p = 1000\ \mathrm{J/(kg\cdot K)}$ - Surface resistances per EN ISO 6946: $R_{\text{si}} = 0.13\ \mathrm{m^2\cdot K/W}$ (horizontal heat flow, walls), $R_{\text{se}} = 0.04\ \mathrm{m^2\cdot K/W}$ (external).

2.3.2 Explicit Internal Thermal Mass Injection (`openubem.idf.builder`)

To prevent numerical instability, realistic European thermal inertia must be embedded. In accordance with EN ISO 52016-1 Table B.14 (five standard classes: Very Light $80\ \mathrm{kJ/(m^2\cdot K)}$ / $14\ \mathrm{Wh}$, Light $110/28$, Medium $165/45$, Heavy $260/78\text{--}87$, Very Heavy $370/105\ \mathrm{Wh/(m^2\cdot K)}$), OpenUBEM injects `InternalMass` objects into every dwelling zone: - Mass surface area: $A_{\text{mass}} = 1.5 \times A_{\text{floor}}$ - Mass material thickness: $d_{\text{mass}} = 0.10\ \mathrm{m}$ (representing interior brick partition / concrete slab) - Thermal capacitance calibration (project values mapped from Table B.14 classes): - Standard European Medium/Heavy: $c_m = 45.0\ \mathrm{Wh/(m^2\cdot K)}$ — Medium class - Spain / Central Europe Heavy: $c_m = 50.0\ \mathrm{Wh/(m^2\cdot K)}$ — between Medium and Heavy - Italy Very Heavy (`IT`): $c_m = 87.0\ \mathrm{Wh/(m^2\cdot K)}$ — Heavy class upper bound - UK Medium-Light (`GB`): $c_m = 32.8\ \mathrm{Wh/(m^2\cdot K)}$ — between Light and Medium

> [!NOTE] The country-specific c_m values are project mapping decisions from the five EN ISO 52016-1 classes, not exact Table B.14 entries. Their provenance must be recorded alongside the archetype parameters.

2.3.3 Fenestration & Total Solar Energy Transmittance (`openubem.idf.surfaces`)

European fenestration standards specify total solar energy transmittance (g_{gl} per EN 410 / ISO 9050) rather than North American Solar Heat Gain Coefficient (SHGC). In EnergyPlus, `WindowMaterial:SimpleGlazingSystem` is parameterized:

$$\mathrm{SHGC} = g_{\text{gl}}, \quad U_{\text{factor}} = U_w$$

The conversion is valid within ± 0.02 for standard residential glazing types. Typical European ranges: standard double clear $g_{\text{gl}} = 0.67\text{--}0.75$; low-emissivity $g_{\text{gl}} = 0.50\text{--}0.60$ (per EN 410:2011).

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3. European Building Typologies & Data Ingestion

3.1 TABULA Building Type Hierarchy

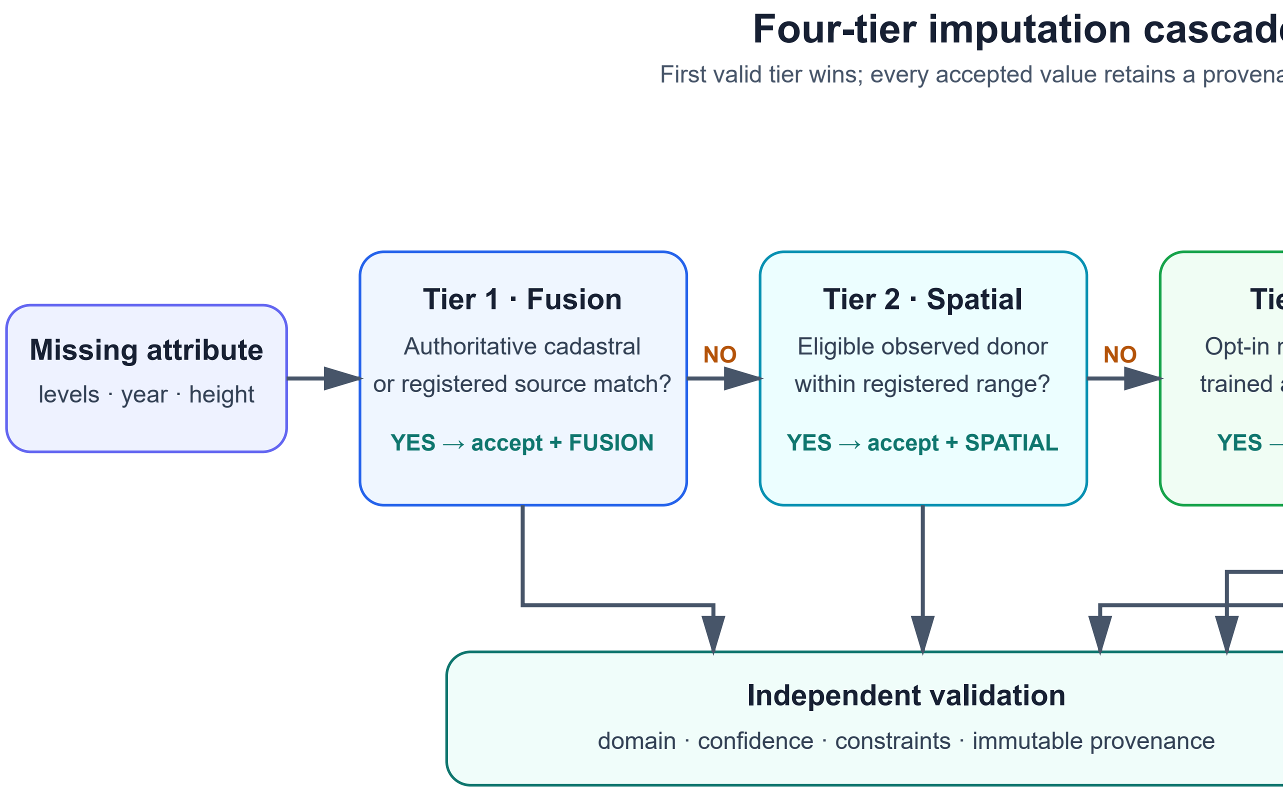
OpenUBEM ingests four standard residential typologies defined in the TABULA / EPISCOPE framework:

Typology	Building type	Typical storeys	Target dwelling layout
`SFH`	Single-family house	1–2	One dwelling zone per storey, with the vertical aggregation rule declared.
`TH`	Terraced house	2–3	One dwelling zone per storey plus paired party walls.
`MFH`	Multi-family house	3–5	Two to four dwellings per storey plus circulation core.
`AB`	Apartment block	4–10+	Four to eight or more dwellings per storey plus corridor/core.

Table 3. Residential typology taxonomy used for ES, GB, IT, and the France physical-model branch. National sources determine the actual period/type combinations; this generic table does not establish country counts. Machine-readable copy: `content/table_3_1_typology_taxonomy.csv` .

3.2 Four-Tier Imputation Cascade for European Footprints

When querying European OpenStreetMap footprints or municipal geospatial portals (e.g. Spanish Catastro, UK Ordnance Survey, Italian Agenzia delle Entrate), missing data attributes are resolved via OpenUBEM's **Four-Tier Imputation Cascade** (`OpenUBEM_imputation_methods.md`):



Four-tier imputation cascade with fusion, spatial, machine-learning, and statistical decisions followed by independent validation or explicit exclusion.

Figure 2. Four-tier imputation cascade. Tiers execute in the current router order—fusion, spatial, opt-in ML, then statistical—and every accepted value is independently checked and labelled with provenance. Failure of all tiers is `BLOCKED/EXCLUDED` *, never a silent default. Reusable sources:* *SVG* *and* *Mermaid* .

Strict Imputation Rules:

- **Zero Fitted Parameters Rule**: Imputation never invents parameters or adjusts coefficients to match an energy target.
- **Minimum Ceiling Height Floor**: All heights are constrained to $h_{\min} \geq 2.10\ \mathrm{m}$ per international building code requirements.
- **Full Traceability**: Every imputed property carries an immutable provenance token (e.g., `HOTDECK_NEIGHBOR_HIGH` , `STATISTICAL_PDE`).

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4. Procedural Floor Layout & Dwelling Slicing Engine

4.1 Algorithmic Geometry Pipeline (from *Iseri et al., 2025* & `kbem_ankara_pipeline.py`)

OpenUBEM's `openubem.geometry.layoutGenerator` implements the 4-stage procedural spatial layout algorithm:


1. GIS Boundary Regularization
Raw 2D Footprint Polygon
ConvexToConcave Transformation
EdgeTo4 Quadrilateral Fitting
UV Domain Reparameterization
(Outward Surface Normals)
2. Vertical Slicing & Storey Zoning
Storey Height Extrusion
(z_k = k * h_storey)
Storey Boundary Tagging
(Ground: Pos 0, Mid: Pos 1, Top: Pos 2)
DataTree Branching: {building_id; floor_id}
3. Floor Layout & Unit Subdivision
Target Flats per Floor (n)
n = 1
n = 2
n = 3, 4
n = 5, 6
n >= 7
1x1 Grid (Full Floor Plate)
2x1 Grid (Dual-Aspect Split)
2x2 Grid (Quad Point-Block)
3x2 Grid (Double-Loaded Spine)
4x2 Grid (High-Density Multi-Family)
4. Circulation & Quality Diagnostics
Embed Centroidal Staircase Core
(6% - 12% of Floor Area)
Boolean Subtraction: Floor - Core
Remainder Units?
N = q * N_flr + r
No
Yes
Branch Remainder Generator (q vs q+1)
Execute Windowless Unit Diagnostic
L_ext = Length(d_Omega_u & d_Omega_ext) >= 2.50m

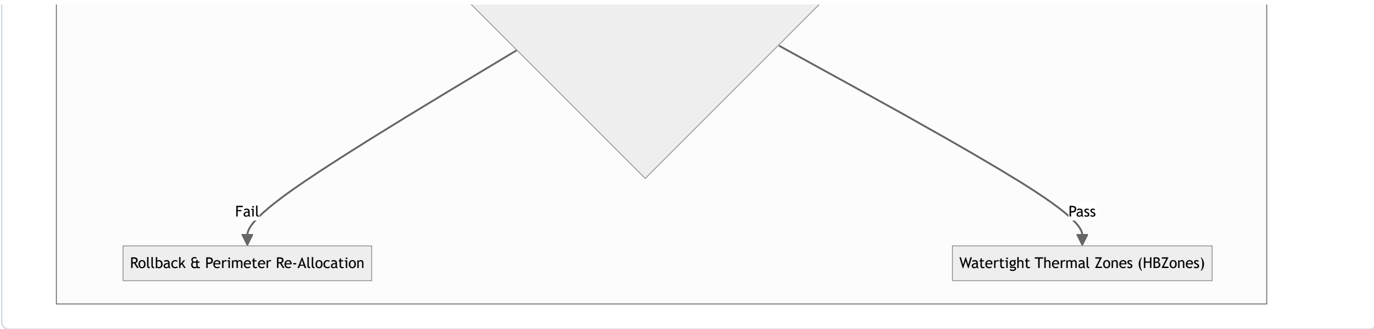
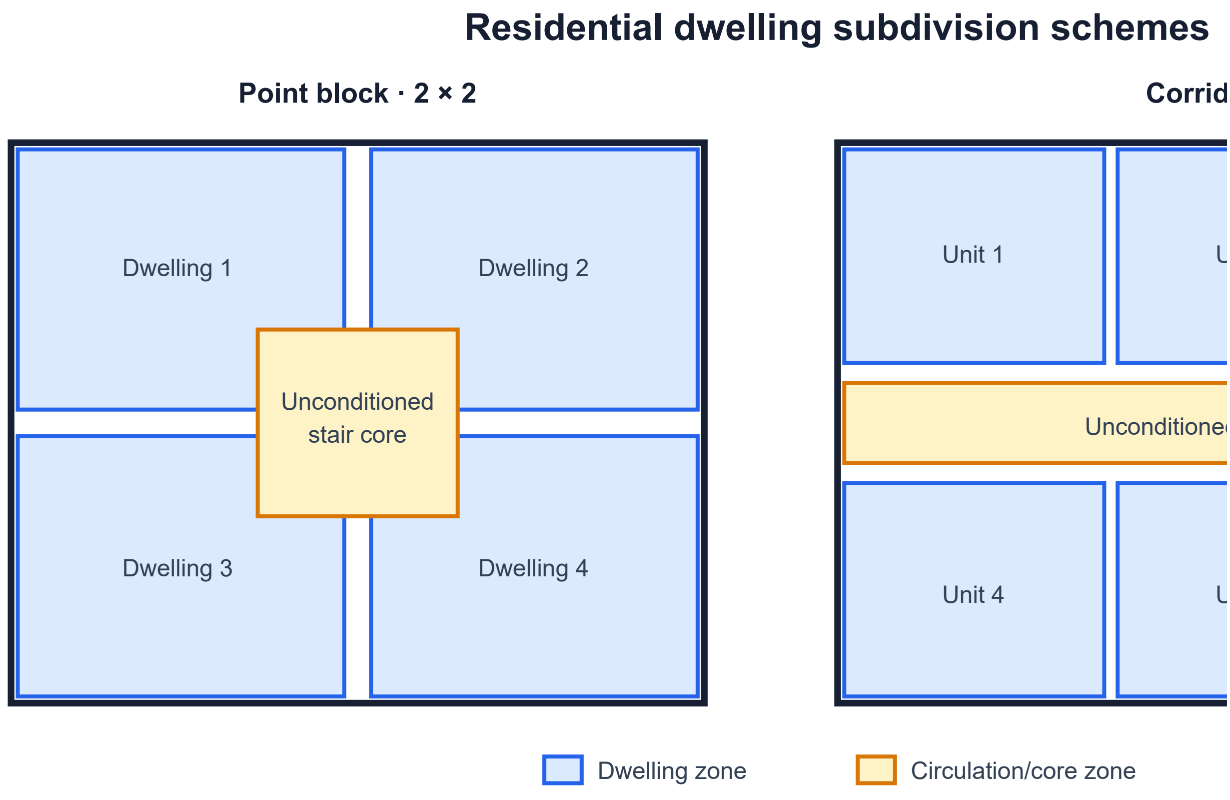


Figure 3. Procedural geometry pipeline from validated residential footprint to watertight dwelling/core zones. Reusable source: [`content/figure_4_1_geometry_pipeline.mmd`](content/figure_4_1_geometry_pipeline.mmd) .

4.2 Slicing Rules & Orthogonal Grid Subdivision



Point-block and double-loaded corridor residential dwelling subdivision schemes.

Figure 4. Conceptual dwelling subdivision schemes. These diagrams express topology, not surveyed room plans; acceptance depends on the tests in Section 4.8. Reusable source: [`content/figure_4_2_dwelling_layout_schemes.svg`](content/figure_4_2_dwelling_layout_schemes.svg) .

Grid Assignment Rules:

- 1×1 **Grid**: Single-Family (`SFH`) / Terraced (`TH`) — 1 thermal zone per floor.
- 2×1 **Grid**: Small Multi-Family — 2 dual-aspect dwelling units per floor.
- 2×2 **Grid**: Point-Block `MFH` — 4 corner quadrant dwelling units surrounding a centroidal stairwell core.
- 3×2 **Grid**: Medium Apartment Block `AB` — 6 dwelling units (4 corner dual-aspect + 2 middle single-aspect) along a central corridor.
- 4×2 **Grid**: Large Apartment Block `AB` — 8 dwelling units along a central double-loaded circulation spine.

4.3 Unconditioned Staircase Core & Buffer Zone Physics

Communal circulation spaces (staircase, elevator shaft, entry vestibule) represent 6%--12% of gross floor area (12.0--$25.0\ \mathrm{m}^2$ per floor; mean $18.4\ \mathrm{m}^2$ in the Ankara validation dataset). - **Zoning Classification**: OpenUBEM models the staircase core as an **explicit unconditioned thermal zone** (`mode: "unconditioned_buffer"`). - **Thermal Behavior**: The staircase zone floats passively ($12.0^{\circ}\mathrm{C}$--$16.0^{\circ}\mathrm{C}$ in winter), buffering heat transfer across party walls between heated apartments and the exterior:

$$b_u = \frac{T_i - T_u}{T_i - T_e} \approx 0.50 \text{ to } 0.80$$

This thermal buffering reduces adjacent dwelling heating demand by 8%--15% (*Iseri et al., 2025*). - **Infiltration Rate**: Staircase zones are assigned a background infiltration rate of $0.000500\ \mathrm{m}^3/(\mathrm{s} \cdot \mathrm{m}^2)$ representing natural air leakage through entry doors and service risers. - **Inter-Zone Surfaces**: Walls separating dwellings from the staircase core are assigned EnergyPlus boundary condition `Surface` linked to the adjacent stair zone. - **Party-Wall Heat Transfer**: Inter-dwelling conduction through shared walls accounts for 15%--40% of net apartment heat loss during unheated or lower-setpoint periods in adjacent flats:

$$Q_{\text{party}} = U_{\text{party}} \cdot A_{\text{party}} \cdot (T_{u1}(t) - T_{u2}(t))$$

4.4 Habitability & Windowless Unit Sanity Gate

To ensure that automated polygon slicing never generates illegal, interior-enclosed dwelling units without exterior access, OpenUBEM enforces the **Windowless Unit Diagnostic Gate**:

$$L_{\text{exterior}} = \text{Length}\left(\partial\Omega_u \cap \partial\Omega_{\text{exterior}}\right) \geq 2.50\ \text{m}$$

If any generated dwelling unit has $L_{\text{exterior}} < 2.50\ \text{m}$, the layout generator rejects the invalid cut and falls back to a conforming single-loaded or dual-aspect subdivision.

4.5 Non-Integer Unit Remainder Stratification

When cadastral records specify a total building unit count N_{units} that is not divisible by storeys N_{floors}:

$$N_{\text{units}} = q \cdot N_{\text{floors}} + r, \quad \text{where } q = \lfloor N_{\text{units}}/N_{\text{floors}} \rfloor, \quad 0 \leq r < N_{\text{floors}}$$

- $N_{\text{floors}} - r$ storeys are partitioned into q units/floor. - r storeys (typically lower floors) are partitioned into $q + 1$ units/floor.

4.6 Geometry Fallback Hierarchy

When a footprint is too narrow, irregular, or degenerate for the target grid, the layout generator applies a deterministic fallback hierarchy: 1. Attempt the assigned grid (2×2, 3×2, etc.) 2. If failed → fall back to the next smaller grid ($2 \times 2 \to 2 \times 1$) 3. If still failed → fall back to 1×1 (full floor plate as single zone)

Every fallback is recorded with an explicit reason token (e.g., `NARROW_FOOTPRINT`, `DEGENERATE_SHAPE`). Fallback must never masquerade as dwelling-level success.

4.7 Empirical Validation Statistics (Ankara KBEM Pipeline)

The procedural layout algorithm was validated on the Ankara KBEM dataset (*Iseri et al., 2025*): - **Sample**: 277 buildings processed, 1,444 floors, 6,458 dwelling units - **Success rate**: 252/277 buildings (91.7%) successfully subdivided into dwelling-level zones - **Fallback rate**: 25 buildings (8.3%) fell back to single-zone-per-floor (causes: narrow footprints $< 8\ \text{m}$ width, L-shaped/highly irregular shapes) - **Mean dwelling area**: $87.3\ \text{m}^2$ (range: 32–$215\ \text{m}^2$) - **Area conservation**: $\leq 0.5\%$ error for all successful subdivisions; zero overlaps detected - **Facade contact**: All dwelling units passed the $2.50\ \text{m}$ exterior threshold

> [!NOTE] The $2.50\ \text{m}$ facade-contact threshold originates from IRC Section R303 and Turkish Zoning Law. It is treated as a declared project modeling rule whose jurisdictional provenance must be confirmed for each European stock (see Section 9.8).

4.8 Verification Protocol and Grasshopper Parity Tasks

Grasshopper is an independent geometric reference, not the acceptance authority. A normalized exchange format—GeoJSON or WKT in a pinned projected CRS—must allow the Grasshopper definition and `openubem.geometry.layoutGenerator` to process the same input footprints. The comparison ignores object ordering and compares topology, zone count, areas, exterior contact, circulation area, and adjacency.

Test	Fixture/sample	Required assertion	Independent comparison or mutation	Acceptance
`GEO-01`	Axis-aligned rectangle	Area conservation and expected dwelling count	Independent polygon union	Error $\leq 1\%$; no gaps/overlaps
`GEO-02`	Rotated rectangle	Orientation invariance	Rotate input and inverse-transform output	Same topology and areas within tolerance
`GEO-03`	L-shaped footprint	Valid non-convex subdivision	Python vs. Grasshopper export	All zones valid or explicit fallback
`GEO-04`	Narrow footprint	Deterministic fallback	Sweep width across the registered threshold	Stable reason token; no false success
`GEO-05`	Courtyard footprint	Hole preservation	Independent topology audit	No dwelling/core crosses the courtyard
`GEO-06`	MFH/AB storey stacks	Reciprocal interzone surfaces	Reopen saved IDF	Exactly one mate for every party face
`GEO-07`	Non-integer dwelling count	Remainder allocation	Independent quotient/remainder calculation	Exact total and deterministic storey assignment
`GEO-08`	Grasshopper golden set	Cross-implementation parity	Normalize and compare both exports	Equal zone count; area/contact within tolerance
`GEO-09`	Corrupted clean fixture	Gate sensitivity	Inject overlap, gap, and unpaired face	Named gate fails; restored fixture passes
`GEO-10`	Residential sample groups	Scale/stability	Run 4, 12, 32, then 96 buildings	Every attempted building accounted for before neighbourhood scale

Table 4. Verification matrix for the procedural floor-layout method, including independent Grasshopper parity and negative controls. Machine-readable copy: [`content/table_4_8_geometry_verification_matrix.csv`].

Each `GEO-*` task must retain the input footprint, generator configuration, normalized Grasshopper export, normalized OpenUBEM export, comparison report, preview image, and test command. Visual similarity alone is not a pass; numerical and topological assertions decide acceptance.

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5. Stochastic Occupancy Injection & 5-Level Sensitivity Sweep

5.1 Mathematical Ingestion Formulation

The occupancy-driven internal gain schedule $\phi_{\text{int}}(t)$ is defined by the pre-registered five-level sensitivity sweep formula:

$$\phi_{\text{int}}(t) = (1 - f) \cdot 3.0 + f \cdot 3.0 \cdot \frac{g(t)}{\text{mean}_{8760}(g(t))}, \quad f \in \{0.00, 0.15, 0.30, 0.50, 1.00\}$$

Level	Factor	Meaning	Campaign status
1	`0.00`	Flat $3.0\ \mathrm{W/m^2}$ control through the final `Schedule:File` path	Required
2	`0.15`	Mild modulation with annual-mean conservation	Required
3	`0.30`	Moderate modulation; not a privileged calibration level	Required
4	`0.50`	High modulation with annual-mean conservation	Required
5	`1.00`	Fully presence-shaped profile normalized to a $3.0\ \mathrm{W/m^2}$ annual mean	Required

Table 5. Five occupant-effect levels for the ES/GB/IT occupant campaign. France uses only the controlled baseline physical path until `FR-OCC-FUTURE` is activated. Machine-readable copy: `content/table_5_1_sensitivity_sweep.csv`.

Fundamental Ingestion Theorems:

1. **Strict Energy Conservation**: $\int_0^{8760} \phi_{\mathrm{int}}(t)\, dt = 3.0 \times 8760 = 26,280\ \mathrm{Wh/m^2}$ for all $f \in \{0.00, 0.15, 0.30, 0.50, 1.00\}$. All observed heating energy variations represent pure temporal load shifting.
2. `Schedule:File` **Ingestion Architecture**: The 8,760 hourly multipliers are written to external CSV files and referenced via EnergyPlus `Schedule:File` objects with `Interpolate to Timestep = No` (enforced per Gate **G8.13**).
3. **Inter-Household Heterogeneity**: In multi-dwelling archetypes (`MFH`, `AB`), each dwelling unit zone $u \in \{1, \dots, N_{\mathrm{units}}\}$ receives an independently sampled demographic presence curve $g_u(t)$ from the held-out LOCO fold population.

5.2 HVAC Plant Modeling Decision (DR07)

Resolving full explicit EnergyPlus hydronic plant loops (boilers, pumps, valves, distribution piping) across 510 multi-zone models introduces severe numerical convergence failures. The campaign standardizes on `ZoneHVAC:IdealLoadsAirSystem` post-processed with natural gas condensing boiler seasonal efficiency curves ($\eta_{\mathrm{seasonal}} = 0.90$, range $0.88\text{--}0.94$), which preserves identical envelope heat balance while achieving 100% convergence across the full campaign matrix.

5.3 Natural Ventilation Assumption

The European campaign maintains a closed-window assumption with continuous background infiltration per TABULA methodology ($n_{\mathrm{air,use}} = 0.30\text{--}0.59\ \mathrm{ACH}$). Natural ventilation window opening is not modeled. This is consistent with the Ankara validation study (*Iseri et al., 2025*) and ensures that the only experimental variable is the occupancy-driven internal gain schedule.

---

6. Simulated vs. Reconstructed EUI Accounting

OpenUBEM enforces the **Simulated vs. Reconstructed EUI Accounting Methodology** (`simulated_vs_reconstructed_methodology.md`):

```
+------------------------------------------------------------------------------------------+
|                                  EUI ACCOUNTING FRAMEWORK                                |
+------------------------------------------------------------------------------------------+
| 1. Simulated EUI (EnergyPlus Physics Output):                                           |
|    EUI_sim = Heating + Cooling + Lighting + Equipment (Plug Loads)                       |
|    - Captures thermodynamic envelope heat balance, solar gains, and dynamic internal gains.|
|                                                                                          |
| 2. Reconstructed EUI (Whole-Building Energy Consumption):                                |
|    EUI_reconstructed = EUI_sim + EUI_service_loads                                       |
|    - Post-processed fraction-split completion adding unsimulated auxiliary loads:       |
|      * Domestic Hot Water (DHW)                                                          |
|      * Cooking & Kitchen Equipment                                                       |
|      * HVAC Distribution Parasitics & Pumps                                              |
|    - Calibrated using TABULA Table 4 national end-use share splits.                      |
+------------------------------------------------------------------------------------------+
```

Figure 5. EUI accounting alternatives. Physical and reconstructed service-load paths are mutually exclusive for any case. Reusable source: `content/figure_6_1_eui_accounting.mmd`.

---

7. Pre-Registered Gate Conformance & Validation Architecture

Every European simulation executed by OpenUBEM is validated against the **Pre-Registered Gate Conformance Matrix**:

Gate	Name	Minimum assertion	Current status
G8.0	Uninjected control	Controls are independently audited before any non-zero release	`NOT_RUN`
G8.8	Scenario differentiation	Registered outputs differ across f levels	`NOT_RUN`
G8.9	Stale-output guard	Any dependency change invalidates cache	`NOT_RUN`
G8.10	Meter tripwire	Component energy reconciles to the selected total	`NOT_RUN`
G8.11	Meter-name validity	Required meters exist and are parseable	`NOT_RUN`
G8.12	Schedule ingestion	Saved IDF references the expected schedule/checksum	`NOT_RUN`
G8.13	Interpolation setting	`Interpolate to Timestep = No`	`NOT_RUN`
G8.14	Manifest completeness	Measured immutable manifest exists per case	`NOT_RUN`
G8.15	Error/warning triage	No fatal/severe error; every warning is classified	`NOT_RUN`
G8.16	Held-out-fold correctness	Country and schedule source satisfy the no-leakage rule	`NOT_RUN`

Table 6. Abbreviated Step 8 gate summary. These are test specifications, not passing results. Machine-readable copy: `content/table_7_1_gate_summary.csv`.

7.1 Negative Control Diagnostic Thresholds (DR05 / DR07)

In addition to the gate matrix, the European campaign employs negative control thresholds to detect gross parameterization errors before the full campaign:

- **Reject** the European standards parameterization if $f = 0.00$ uninjected control runs produce heating EUIs deviating by $> 50\%$ from published TABULA national brochure benchmarks ($Q_H \approx 80\text{--}180\ \mathrm{kWh/m^2a}$).
- **Reject** the adapted OpenUBEM configuration if median EUI at $f = 0.00$ differs from TABULA national baseline by $> 25\%$ (DR07 negative control).
- **Reject** if pre-1945 uninsulated Spanish/Italian archetypes at $f = 0.00$ yield space heating $< 50\ \mathrm{kWh/m^2a}$ (DR06 negative control).

These are diagnostic flags during Q1–Q2 qualification stages (Section 10), not calibration targets. The pipeline must never alter a TABULA parameter to make a pilot EUI appear more plausible.

---

8. Clean Repository Layout & Module Specifications

```
C:\Users\o_iseri\Desktop\OpenUBEM\
├── docs/
│   ├── docs_EXPLANATION/               <- Core Methodology & Reference Guides
│   │   ├── OpenUBEM_fundamentals.md
│   │   ├── OpenUBEM_inputs_reference.md
│   │   ├── OpenUBEM_imputation_methods.md
│   │   ├── simulated_vs_reconstructed_methodology.md
│   │   └── OpenUBEM_debug_References.md (~200 Error Patterns Registry)
│   └── docs_ACTIVE/
│       └── europeanLocations/          <- Active European Implementation & Walkthroughs
│           ├── MVP_european_locations.md (This Implementation Specification File)
│           └── WALKTHROUGH_european_locations.md (Operational Guide & Walkthrough)
└── openubem/
    ├── acquisition/
    │   ├── osm_fetcher.py              -> Ingests OSM European footprints & tags
    │   ├── overture_fetcher.py         -> Tier 1 height/levels fusion
    │   └── epw_manager.py              -> Handles Madrid, London, Bologna AMY weather files
    ├── data/
    │   ├── construction/
    │   │   ├── tabula_archetypes_es.json -> Spanish envelope assemblies & U-values
    │   │   ├── tabula_archetypes_uk.json -> UK Part L envelope assemblies & U-values
    │   │   └── tabula_archetypes_it.json -> Italian DM 26/06/2015 assemblies & U-values
    │   ├── loads/
    │   │   └── european_residential_loads.json -> 3.0 W/m² base internal load densities
    │   └── schedules/
    │       └── tabula_statutory_schedules.json -> Normative flat European schedules
    ├── geometry/
    │   ├── footprint.py                -> Footprint cleaning, UTM reprojection, metrics
    │   ├── layoutGenerator.py          -> Procedural dwelling subdivision & unconditioned stair
    │   └── zoning.py                   -> Adaptive zoning strategy decision engine
    ├── idf/
    │   ├── builder.py                  -> Watertight IDF assembly & InternalMass injection
    │   ├── opaque_assembly.py          -> Parametric insulation thickness sizing from U-values
    │   ├── surfaces.py                 -> SimpleGlazingSystem total solar transmittance (g_gl)
    │   └── hvac.py                     -> Hydronic radiator convective plant / IdealLoads curves
    ├── semantic/
    │   ├── building_classifier.py      -> Maps OSM tags to TABULA archetypes (SFH, TH, MFH, AB)
    │   ├── construction_sets.py        -> Resolves vintage standards across 22 epochs
    │   ├── imputation.py               -> 4-tier zero-fitted imputation cascade
    │   └── schedules.py                -> Schedule:File generation (Interpolate to Timestep = No)
    ├── simulation/
    │   ├── runner.py                   -> Local EnergyPlus execution & meter verification
    │   └── parallel.py                 -> SLURM cluster array runner for 510 campaign cells
    └── results/
        ├── service_loads.py            -> Fraction-split EUI completion (simulated vs reconstructed)
        └── carbon.py                   -> National grid carbon factors (ES, UK, IT)
```

---

9. v1.1 Code-Audited Implementation Addendum (Authoritative Delta)

9.1 Purpose, Vocabulary, and Source Precedence

This addendum turns the preceding target architecture into an implementable MVP contract while preserving the original proposal. The following status vocabulary is mandatory in issues, manifests, pull requests, and reports:

- **CURRENT**: present in the OpenUBEM repository and exercised by an existing test.
- **REUSABLE**: present, but requiring a European adapter or new configuration.
- **TARGET**: specified here but not yet implemented.
- **BLOCKED**: cannot be accepted until a named upstream artefact or decision exists.
- **VERIFIED**: supported by an artefact produced by an executed test or simulation; prose alone is not verification.

When two documents disagree, use this precedence order:

1. `Step8_docs/4thJ_08_bemSimulation.md` and `Step8_docs/4thJ_08_bemSimulation_val.md`, including their dated rulings;
2. measured repository code and tests at the commit recorded in the campaign manifest;
3. `IMP_step8/4thJ_08_bemSimulation_IMP.md`;
4. `IMP_step8/outputs/` and `IMP_step8/DeepResearch/` synthesis documents;
5. illustrative examples in Sections 1–8 above.

This ordering matters because some synthesis outputs predate later rulings. In particular, any document that reports **612 executions** or **510 injected runs plus 102 controls** is superseded by the active ruling that makes `f=0` the control endpoint of the five-level sweep.

9.2 Repository Baseline Audit (2026-08-22)

Capability	Status	Evidence in OpenUBEM `0.1.0`	MVP delta
OSM acquisition and projected polygon cleaning	**CURRENT**	`openubem.acquisition.osm_fetcher.ingest_buildings` (`fetch_buildings` alias)	Add European test fixtures; do not create a second fetcher API.
Four-tier imputation router	**CURRENT/REUSABLE**	`openubem.semantic.imputation.impute_missing(gdf, cfg=None, targets=None, rng=None)`	Add country-aware donors/configuration; retain existing provenance column naming `provenance_<attribute>`.
Room-layout geometry	**CURRENT/REUSABLE**	`openubem.geometry.layoutGenerator.generate_layout(...)`	Existing specifications are DOE/IBC-based and recognize OpenUBEM archetype IDs such as `MidriseApartment`; add explicit TABULA residential module specifications and dwelling/core semantics.
IDF generation	**CURRENT/REUSABLE**	`openubem.idf.builder.BuildingIDF` and `run_step3(...)`	Extend the existing builder. The proposed `IDFModelBuilder` API in the walkthrough does not currently exist.
Opaque U-value realization	**CURRENT/REUSABLE**	`build_opaque_assembly(idf, name, u_value, thermal_mass)`	Current assembly is a generic `_K=0.12 W/m·K` resistance/mass representation, not the proposed brick–EPS–plaster construction. Add a traceable European construction path without regressing the CTF stability cap.
Schedule generation	**CURRENT, incompatible with Step 8**	`openubem.semantic.schedules` writes six DOE `Schedule:Compact` objects	Add an external-file schedule adapter and saved-IDF read-back. No current function writes the required Step 8 `Schedule:File`.
HVAC assignment	**CURRENT/REUSABLE**	`openubem.idf.hvac.assign_hvac` supports current OpenUBEM archetypes	Add and validate European residential systems. Do not describe hydronic radiators as implemented until emitted objects and meters are tested.
Simulation fan-out	**CURRENT/REUSABLE**	`openubem.simulation.parallel.run_neighbourhood` uses joblib and writes `04_simulation_manifest.parquet`	Add a campaign-cell wrapper and a separate SLURM submission script. `parallel.py` is not currently a SLURM array CLI.
Results reconstruction	**CURRENT, US-specific/default-off**	`openubem.results.service_loads.reconstruct_frame`; reconstruction is disabled by default because Phase E physically models five service loads	Add European coefficients only if the campaign explicitly chooses reconstruction mode; prevent double counting. The proposed `reconstruct_eui` function does not exist.
Step 8 gates	**TARGET**	Existing `compute_validation_gates` is CBECS-oriented, not a G8.0–G8.16 scorer	Implement a dedicated Step 8 scorer, mutation tests, and vacuity guards. No G8 gate is currently verified.
European TABULA files	**TARGET**	`tabula_archetypes_{es,gb,it,fr}.json`, `european_residential_loads.json`, and `tabula_statutory_schedules.json` are absent	Generate them from pinned TABULA sources, with row-level provenance and licence record; keep the FR physical registry separate from the 102-record occupant registry.

Table 7. Code-audited repository capability baseline at the stated review date. Machine-readable copy: [`content/table_9_2_repository_baseline.csv`](content/table_9_2_repository_baseline.csv).

9.3 Frozen Scientific Decisions and Explicit Non-Decisions

The MVP must encode the following active decisions without silently substituting an earlier proposal:

Topic	Frozen MVP contract
National populations	Spain (`es`), England-labelled TABULA stock (`gb` , while the survey fold may remain `uk`), and Italy (`it`). Do not describe TABULA `GB` as the whole United Kingdom without the England limitation.
France physical-model scope	France (`fr`) is included in residential acquisition, enrichment, layout, IDF, weather, and baseline physical-simulation work. Its registry count must be audited before any combined physical-baseline total is published.
France occupant scope	France-specific diaries, held-out-fold logic, and non-zero occupant schedules are `FUTURE_SCOPE` . French controlled baselines remain separate from the 102-archetype/510-run ES/GB/IT occupant campaign.
Archetype population	24 ES + 36 GB + 42 IT = **102 archetypes**. Each country is simulated only with the model fold that held that country out.
Sensitivity grid	`f ∈ {0.00, 0.15, 0.30, 0.50, 1.00}` . Report every level; do not designate `f=0.30` as a primary or calibrated level.
Campaign size	**510 annual runs per weather specification**: 102 archetypes × 5 levels. `f=0` is included and must run/read first for each matching archetype/weather configuration.
Gain baseline	Use the TABULA EU boundary-condition baseline of exactly `3.0 W/m²` in all three folds. Italian `4.0 W/m²` is contextual information, not the Step 8 campaign baseline.
Conservation	Every valid `phi_int` series has annual mean `3.0 W/m²` , within a documented numerical tolerance. Differences across `f` redistribute gains in time; they do not change annual internal-gain energy.
Zoning	One thermal zone per dwelling. A dwelling may have a non-shoebox exterior geometry, but Step 7 provides `at_home` rather than room-level location, so the study makes no within-dwelling spatial claim.
Household assignment	Independent diary assignment per dwelling is a **target sampling rule** that must be seeded, recorded, and tested; it is not yet current OpenUBEM behavior.
Weather	Use actual meteorological weather covering each diary-survey fieldwork window. The location, exact twelve-month selection, source, licence, and checksum remain required acquisition records. Compare the `f` effect within fold; any absolute cross-fold comparison must name the meteorological year.
Heating intermittency	Preserve the TABULA reduction factor as a scalar. Do not add a thermostat night-setback schedule that would confound the occupancy treatment.
Geometry and constructions	Do not claim that TABULA supplies floor layouts, aspect ratio, orientation, window-to-face placement, or material layer build-ups. Those assumptions require separate provenance and sensitivity treatment.

Table 8. Frozen decisions and explicit non-decisions for the implementation. The France physical branch is current scope while the France occupant branch is deferred. Machine-readable copy: [`content/table_9_3_frozen_decisions.csv`](content/table_9_3_frozen_decisions.csv) .

9.4 Ownership and Boundary Contract

OpenUBEM owns reusable building-physics capabilities:

- TABULA parameter loading and schema validation;
- country/archetype translation into existing semantic rows;
- European construction, glazing, HVAC, and weather adapters;
- dwelling/core geometry and watertight IDF generation;
- generic external schedule-file emission and saved-IDF inspection;
- EnergyPlus execution, parsing, and low-level meter integrity.

GSSCanada owns experiment-specific information:

- held-out LOCO fold selection and Step 7 schedule provenance;
- diary chaining and household sampling rules;
- the five-level campaign matrix and run ordering;
- Step 8 `manifest.json` , gate scoring, mutation probes, and scientific reporting.

The coupling boundary is a versioned, immutable **campaign-cell specification**. OpenUBEM must not infer the held-out fold from the country filename, and GSSCanada must not reach into private OpenUBEM geometry or IDF internals to patch objects after generation.

9.5 Required Input Contracts

TABULA archetype record

Every one of the 102 records must include, at minimum:

```
archetype_id, country_stock_code, survey_fold, construction_period,
building_type, source_workbook_sha256, source_sheet, source_row,
u_wall_w_m2k, u_roof_w_m2k, u_floor_w_m2k, u_window_w_m2k,
g_gl_window, n_air_use_h_1, phi_int_w_m2, c_m_wh_m2k,
f_red_htr, parameter_assumptions
```

Requirements:

- units are encoded in field names and checked at load time;
- source cells and transformations are recorded, not only the workbook name;
- refurbishment variants and unclassified rows are excluded with an explicit reason table;
- country stock code (`GB`) and survey fold (`uk`) are separate fields;

- missing values are never replaced by an energy-calibrated parameter.

Step 7 presence-series record

Each `g(t)` artefact must carry:

```
schedule_id, schedule_path, schedule_sha256, fold, held_out_country,
diary_source_id, chaining_rule, timezone, local_time_basis,
n_hours, start_timestamp, end_timestamp, random_seed
```

The adapter must reject a series when it contains non-finite or negative values, has `mean(g) <= 0`, has a length inconsistent with the selected EnergyPlus run period, or names the wrong held-out fold. Daylight-saving and leap-hour treatment must be resolved once in the campaign builder and recorded; silent truncation or duplication is forbidden.

Weather record

Each EPW/AMY record must include location, covered dates, source URL or archive identifier, licence, acquisition date, SHA-256, parsed EPW location, and the rule connecting its period to the diary fieldwork window. OpenUBEM's builder already fails if `epw_path` is missing and populates `Site:Location` from the file; the European adapter must retain that fail-fast behavior.

9.6 Campaign-Cell and Manifest Contract

A deterministic cell identifier should be constructed from normalized dimensions, for example:

```
<country_stock_code>__<archetype_id>__<weather_id>__f<000|015|030|050|100>
```

Each cell manifest must be written atomically and include at least:

```
{
  "schema_version": "step8-cell-manifest/1.0",
  "cell_id": "GB__GB.ENG.MFH.05__uk_amy_2014_2015__f030",
  "archetype_id": "GB.ENG.MFH.05",
  "country_stock_code": "GB",
  "held_out_country": "uk",
  "fold": "uk_held_out",
  "sensitivity_f": 0.30,
  "schedule_source_sha256": "<measured>",
  "schedule_emitted_sha256": "<measured>",
  "idf_sha256": "<measured>",
  "weather_sha256": "<measured>",
  "openubem_version": "0.1.0",
  "openubem_git_commit": "<measured>",
  "energyplus_version": "<measured>",
  "energyplus_build_hash": "<measured>",
  "platform": "<measured-at-runtime>",
  "random_seed": 42,
  "created_utc": "<measured>",
  "status": "planned|running|success|failed"
}
```

Checksums must be computed from files on disk. A cache hit is valid only when a dependency digest covering the IDF, emitted schedule, weather, engine build, adapter configuration, and relevant source commit is unchanged. The existing OpenUBEM `is_completed()` check based only on `.end` and `.sql` is insufficient for G8.9 and must be wrapped rather than weakened.

9.7 Implementation Work Packages

WP	Deliverable	Primary modules	Acceptance evidence
EU-01	Versioned TABULA loader: frozen 102-record ES/GB/IT occupant registry plus audited FR physical registry	`openubem/data/construction/`, `openubem/semantic/construction_sets.py`	Schema tests; 24/36/42 counts; separate FR count/provenance report; deterministic regeneration.
EU-02	European semantic crosswalk and residential-use filter	`building_classifier.py`, `construction_sets.py`	Year-boundary tests; explicit `GB` / `uk`; FR physical mapping; all non-residential/unknown exclusions counted; no US fallback without a flag.
EU-03	European envelope and mass adapter	`opaque_assembly.py`, `builder.py`, `surfaces.py`	U-value tolerance, `g_gl → SHGC`, CTF stability, and saved-IDF construction readback on S0–S3 samples.
EU-04	Dwelling/core layout adapter	`layoutGenerator.py`, `zoning.py`, `surfaces.py`	`GEO-01` – `GEO-10`; Grasshopper parity; S0–S3 sample groups; area conservation ≤1%; no overlaps/gaps; facade access; paired surfaces; deterministic fallback.
EU-05	Residential HVAC/ventilation adapter	`hvac.py`, builder	Object tests and autosizing/design-day smoke tests on sampled ES/GB/IT/FR dwellings; fuel/end-use meters; no conditioning in circulation core.
EU-06	External occupancy schedule adapter	`semantic/schedules.py`, builder	Five conserved series; `Schedule:File`; `Interpolate to Timestep=No`; saved-IDF independent read-back; correct `People`/gain assignment.
EU-07	Weather registry	`acquisition/epw_manager.py`	Period/location/licence/checksum records; baseline FR weather; fieldwork-aligned ES/GB/IT occupant weather; no cross-cell drift within a specification.
EU-08	Residential campaign and SLURM wrapper	new GSSCanada driver plus `simulation.parallel` reuse	S0–S3 pass before neighbourhood scale; 510-row ES/GB/IT occupant manifest; separate FR baseline manifest; resumable dependency-hash cache.
EU-09	Step 8 scorer and mutation suite	new Step 8 validation module	G8.0–G8.16, V8.a–V8.g, all mandated perturbations seen failing, null perturbation stays clean.
EU-10	Results and dossier export	results adapter	Annual/monthly/hourly/peak outputs; explicit EUI accounting mode; no duplicated service loads; machine-readable gate report.

Table 9. EU-01 through EU-10 implementation work packages. Machine-readable copy: [`content/table_9_7_work_packages.csv`].

9.7.1 Residential-Only Sample-Group Qualification

The geospatial acquisition layer may retain all source footprints for audit counts, but the active construction and simulation registry must include **residential buildings only**. Non-residential and unknown-use buildings receive an explicit exclusion record before layout generation; they are not imputed into a residential typology and are not simulated.

Stage	Residential buildings	Purpose	Simulation scope	Promotion rule
`S0`	4 synthetic fixtures	One footprint per SFH, TH, MFH, AB	Geometry and IDF construction only	All geometry and saved-IDF gates pass
`S1`	12 observed buildings	Three per typology; simple and irregular footprints	Short design-day smoke simulations	12/12 accounted for; failures classified
`S2`	32 observed buildings	Old/new and high/low data-completeness strata	Short-period simulations	Stable outputs and measured resources
`S3`	96 observed buildings	Balanced multi-country residential pilot, including FR physical cases	Annual controlled-baseline simulations	Approved exclusions and measured resource envelope
`N1`	500–600 residential buildings	First neighbourhood-scale study after S0–S3	Staged controls before any occupant cases	Independent input and control audits pass
`N2`	Up to 1,000 residential buildings	Optional scale-up after N1	Same manifest/dependency contract	Measured capacity and explicit approval

Table 10. Residential-only sample-group ladder. Counts are target sample sizes, not evidence that a dataset or simulation already exists. Machine-readable copy: [`content/table_9_7_sample_group_ladder.csv`].

No full neighbourhood is built or simulated merely because a single synthetic case succeeds. Each promotion must retain per-building geometry, IDF, warning, meter, runtime, memory, and exclusion evidence.

9.8 Geometry Acceptance Rules

The Ankara/Grasshopper artefacts are methodological references, not drop-in production code. The European adapter must satisfy repository-level invariants for every generated floor:

- valid, finite, projected polygons with a consistent orientation;
- union of dwelling and circulation polygons equals the cleaned floor plate within **1% area tolerance**;
- no overlap with positive area and no unintended gap;
- every conditioned dwelling has exterior-facade contact, with any `2.50 m` threshold treated as a project modeling rule whose jurisdictional provenance is recorded, not as a universal European code;

- each party wall has a reciprocal EnergyPlus `Surface` partner with matching vertices;
- corridors/stairs are tagged separately and receive no `People`, dwelling equipment, or active thermostat unless explicitly justified;
- unsupported or degenerate shapes fall back deterministically and record the reason; fallback must never masquerade as dwelling-level success.

Because the active Step 8 ruling requires **one zone per dwelling**, a commercial core/perimeter result or a generic room-layout result is not acceptable merely because it is watertight.

9.9 Internal-Gain Semantics

The quantity `phi_int(t)` is a power density in `W/m²`, not an occupancy fraction. It must therefore drive a dedicated internal-gain object or a normalized schedule multiplied by a fixed design level—never a `Fractional` schedule containing values greater than `1.0`.

For each diary:

```
g_norm(t)   = g(t) / mean(g)
phi_int(t)  = 3.0 * ((1 - f) + f * g_norm(t))
```

The implementation must define which EnergyPlus objects receive the combined gain and how radiant/latent/convective fractions are handled. It must also prevent the existing DOE occupancy, lighting, and equipment schedules from remaining active in parallel unless that composition is explicitly part of the experiment. An object-assignment audit is required because matching schedule values alone cannot detect a `People` or equipment object pointing to the wrong schedule.

9.10 EUI Accounting Decision Gate

The original four-end-use reconstruction approach conflicts with the current OpenUBEM Phase E default, which physically emits DHW, cooking, refrigeration, and elevators and disables reporting-layer uplift. Before EU-10 is implemented, the campaign owner must choose exactly one mode:

1. **Physical mode (preferred when the emitted European systems are validated):** simulate the supported service loads and set reconstruction off; or
2. **Four-end-use mode:** disable all corresponding physical service-load objects and apply a European, traceable reconstruction table in post-processing.

Mixing the two modes double-counts energy and is an automatic campaign failure. The manifest must record `eui_accounting_mode`, modeled end uses, reconstructed end uses, denominator area, and coefficient-table checksum.

9.11 Full Validation Contract

The abbreviated table in Section 7 is not the complete gate suite. The MVP scorer must implement:

- **G8.0:** read the matching `f=0` control before quoting `f>0`;
- **G8.1–G8.4:** reproducibility gates against a re-run of the same cell, with thresholds unchanged (monthly/hourly NMBE and CV(RMSE)); these are not measured-accuracy claims;
- **G8.5–G8.6:** peak magnitude and timing checks against the named comparison series;
- **G8.7:** as-modeled published band is graded, empirical band is informational;
- **G8.8–G8.16:** scenario differentiation, stale-cache invalidation, meter balance, meter-name validity, independent saved-IDF schedule ingestion and assignment, interpolation, manifest completeness, warning/convergence triage, and held-out-fold correctness.

It must also implement vacuity guards **V8.a–V8.g** from the pre-registered validation document and the complete perturbation matrix. A gate is not accepted merely because it passes clean data: every gate must be observed failing its designated mutation, while the null perturbation fails none. Warnings are grouped and adjudicated by kind, not hidden by frequency.

9.12 MVP Definition of Done

The European-location MVP is complete only when all of the following are true:

1. The 102-record TABULA registry regenerates deterministically from pinned sources and passes schema/provenance checks.
2. All unresolved design assumptions—geometry, layer build-up, archetype selection, weather location/period/licence, and EUI accounting mode—are written and approved.
3. A representative smoke matrix covers all three stocks, SFH/TH/MFH/AB, old/new construction, simple/complex footprints, all five `f` values, and each selected weather file.
4. Every smoke IDF has valid constructions, one zone per dwelling, correct unconditioned-core behavior where applicable, correct schedule assignment, and zero severe/fatal errors.
5. Schedule conservation, fold identity, disk checksums, saved-IDF read-back, and dependency-based cache invalidation pass.
6. The scorer consumes the declared campaign table, passes all vacuity guards, and its mutation suite demonstrates every gate firing.
7. The full manifest declares exactly **510 expected cells per weather specification**, exactly 102 at each `f`, no duplicate `cell_id`, and no injected result is reported before its matching `f=0` result.
8. Results clearly separate simulated from reconstructed end uses, name the denominator, state the weather year in cross-fold absolute comparisons, and never label a design target as verified evidence.

Until these conditions hold, the correct project status is **implementation plan reviewed; European Step 8 campaign not yet verified**.

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10. Speed HPC and Pre-Occupant Simulation Qualification Addendum

10.1 References and Scope

This section adds an optional high-throughput execution path using Concordia's Speed cluster and a mandatory simulation qualification stage before stochastic occupant information is introduced. It complements:

- the [Speed HPC manual] (version 7.5 at the 2026-08-22 review);
- the local [Parallel IDF Preparation — Technical Reference];
- `scripts/cluster/README.md` and the existing `submit_fleet*.sbatch` templates.

The Speed manual describes CPU batch jobs on the `ps` partition, SLURM job arrays, a maximum batch duration of seven days, and automatic cleanup of inactive `/speed-scratch` files after 90 days. These are operational constraints, not permission to consume every visible CPU. Current queue state, account limits, and fair-share policy remain authoritative at submission time.

10.2 Parallelism Model: More Than 32 CPUs Without Oversubscription

EnergyPlus building simulations are independent and effectively single-core in the existing OpenUBEM workflow. Therefore, parallelism scales across campaign cells:

```
one SLURM array task = one annual EnergyPlus cell = one CPU
total concurrent CPUs ≈ number of simultaneously running array tasks
```

Do **not** request `--cpus-per-task=32` or `64` for one EnergyPlus cell. To use more than 32 CPUs concurrently, keep `--cpus-per-task=1` and raise the array throttle so SLURM can distribute independent cells over multiple nodes.

Profile	Array throttle	Maximum concurrent E+ cells	Intended use	Authorization
Diagnostic	`%4`	4	Four-country physical smoke and debugger-friendly logs	Default
Pilot	`%8` or `%16`	8–16	Representative target 32-case physical qualification matrix	Default
Standard production	`%32`	32	Established OpenUBEM full-fleet policy	Default ceiling
Expanded production	`%48` or `%64`	48–64	Step 8 controls or injected campaign when Speed has capacity	Explicit campaign-owner approval after pilot
Exceptional burst	`> %64`	Scheduler-dependent	Only when justified by measured runtime/memory and approved by Speed/account management	Not a default option

Table 11. Speed execution profiles. A throttle is an upper bound, and expanded profiles require measured pilot evidence. Machine-readable copy: [*content/table_10_2_execution_profiles.csv*](content/table_10_2_execution_profiles.csv).

The throttle is an upper bound, not a reservation. SLURM may run fewer tasks according to availability and fair share. The campaign manifest records both the requested throttle and the measured `AllocCPUS`/elapsed state from `sacct`.

At the current conservative request of `6G` per task, `%64` advertises as much as `384G` of aggregate memory across the cluster. The request must be reduced only after pilot evidence from `MaxRSS`/`MaxVMSize`; it must never be reduced merely to force more concurrency.

10.3 Mandatory Pre-Occupant Qualification Ladder

No `f>0` GSSCanada schedule may run until the following ladder passes. These tests use the final European physics and a constant statutory gain of `3.0 W/m²`; they do not use a stochastic Step 7 diary.

Stage	Population	Parallel option	Purpose	Promotion rule
Q0 — Local unit/sample tests	S0–S2; no annual fleet simulation	Local pytest	Registry parsing, residential filtering, geometry/Grasshopper parity, schedule emission, manifest, saved-IDF read-back	All hard tests pass and every sample is accounted for
Q1 — Four-country physical smoke	4 controlled cases: ES, GB, IT, FR	Local or Speed `%4`	Prove environment, EPW, IDD, ExpandObjects, EnergyPlus, parser, and output retention	4/4 success; zero severe/fatal
Q2 — Stratified physics pilot	Target 32 cases: 4 stocks × 4 residential types × 2 old/new bands	Speed `%8` or `%16`	Exercise envelope, dwelling/core geometry, HVAC, meters, representative weather, and France physical branch	32/32 success or every exclusion approved before proceeding
Q3 — ES/GB/IT full control campaign	102 cases at `f=0`	Speed `%32` default; `%64` expanded	Establish the matched control endpoint for the occupant campaign	G8.0 evidence complete and reviewed
FR-B — France baseline campaign	One controlled case per accepted FR physical archetype	Separate manifest; throttle set from Q2 measurements	Establish France physical baselines without occupant schedules	All FR cases accounted for; no merge into the 510-case denominator
Q4 — Occupant-modulated campaign	408 cases at `f∈{0.15,0.30,0.50,1.00}`	Speed `%32` or approved `%64`	Measure the temporal occupant effect	Submitted only after the Q3 audit job succeeds

The Q3 and FR-B controlled cells must use the same final `Schedule:File` emission and IDF assignment path as later occupant-enabled cases, but the emitted series is constant and requires no occupant diary values. This tests the coupling mechanism without allowing demographic information to affect the physics baseline.

Table 12. Qualification ladder for residential physical models and the ES/GB/IT occupant campaign. France participates through FR-B; France occupant cases remain deferred. Machine-readable copy: [*content/table_10_3_qualification_ladder.csv*](content/table_10_3_qualification_ladder.csv).

10.4 Input-Audit Maps Before Simulation

Before Q1 or Q2, generate a four-panel spatial audit figure patterned after `IMP_step8/resources/Screenshot_3.png`:

1. **Construction period / TABULA band** — categorical, with missing/unclassified buildings visible;
2. **EPC availability** — `EPC available`, `EPC unavailable`, and `not applicable/unknown` as distinct states;
3. **Building function / residential typology** — SFH, TH, MFH, AB, plus non-residential and unknown classes shown only as excluded audit categories;
4. **Construction material / assigned construction set** — observed material where available and the assigned TABULA construction family, with provenance encoded separately.

The map package must include the plotted GeoPackage/Parquet source and a category-count CSV. A visually plausible map is insufficient: source counts, explicit exclusions, and residential simulation-registry counts must reconcile exactly. Non-residential and unknown-use footprints may remain grey/hatch-marked in the audit view, but they must be absent from geometry, IDF, and simulation manifests. Recommended additional panels are imputation provenance/confidence, number of storeys, dwelling count, geometry fallback status, and selected weather ID.

Illustrative axonometric of a large residential study domain with SFH, TH, MFH, and AB buildings inside the boundary and non-residential footprints excluded outside it.

Figure 6. Illustrative 500–1,000-building residential neighbourhood concept. Colours distinguish the four TABULA morphology classes; grey hatched footprints are excluded non-residential context. This is a planning illustration—not a generated dataset, exact building count, input-audit map, or simulation result. Reusable asset: `content/figure_neighbourhood_residential_typologies.png` .

10.5 Pre-Occupant Acceptance Gates

The qualification report must demonstrate:

- **Input completeness:** no silent unknown stock code, period, typology, construction set, EPW, or denominator area;
- **Spatial plausibility:** mapped categories agree with the underlying table and no join silently drops a building;
- **Geometry:** valid zones, ≤1% area drift, no positive-area overlaps/gaps, reciprocal interzone surfaces, and explicit fallback reasons;
- **Physics:** target U-values and glazing parameters are realized within the registered tolerance; circulation zones remain unconditioned;
- **Schedule baseline:** constant `3.0 W/m²`, correct duration/time basis, `Interpolate to Timestep = No`, and independent saved-IDF assignment verification;
- **Execution:** correct EnergyPlus version/build, zero severe/fatal errors, and warning kinds adjudicated;
- **Meters:** every requested meter exists and the G8.10 energy-balance residual is within `0.5%` ;
- **Results:** finite non-negative annual end uses, correct floor-area denominator, and no duplicate/double-counted service load;
- **Reproducibility:** a repeated Q1 cell satisfies G8.1–G8.4;
- **HPC integrity:** expected and observed array-task counts match, no missing outputs, and local checksums match harvested Speed files.

EUI plausibility ranges in this phase are diagnostic flags, not calibration targets. The pipeline must never alter a TABULA parameter to make a pilot EUI appear more plausible.

10.6 Dependency-Enforced Campaign Order

The Speed campaign is submitted as separate, auditable jobs:

```
Q3 control array (102 × f=0)
        │ afterok
        ▼
control harvest + G8.0 audit job
        │ afterok
        ▼
Q4 injected array (408 × f>0)
        │ afterok
        ▼
harvest + full G8/V8 audit
```

Figure 7. Dependency-enforced ES/GB/IT occupant campaign order. The France physical-baseline branch runs under a separate manifest and does not unlock France occupant cases. Reusable source: `content/figure_10_6_dependency_chain.mmd` .

This dependency chain is stronger than placing all 510 rows in one array: an array scheduler may start `f>0` tasks before all controls finish. The control-audit job must exit non-zero if any Q3 cell or gate is missing, preventing the injected array from becoming eligible.

10.7 Step 8-Specific Speed Requirements

The generic OpenUBEM cluster templates are reusable but not sufficient unchanged:

- the Step 8 task list must identify `cell_id`, IDF, **explicit EPW**, gain CSV, and manifest path per row; selecting the first `*.epw` in a directory is forbidden when multiple weather files exist;
- retain `eplusout.sql`, `.err`, `.end`, `.eio`, meter dictionary evidence (`.mdd` or an equivalent archived preflight), saved IDF, emitted gain CSV, `task.rc`, and the runtime manifest until validation completes;
- capture non-zero EnergyPlus return codes even when the shell uses `set -e`;
- use per-cell output directories and `%A_%a` logs to prevent collisions;
- write to `/speed-scratch/$USER/...`, use node-local `$TMPDIR` for repeated temporary I/O where practical, and copy all durable results off Speed promptly;
- use the CPU `ps` partition; GPU partitions provide no benefit for EnergyPlus;
- never run EnergyPlus on the login node.

10.8 HPC Evidence and Definition of Done

Each array submission adds an HPC record containing:

```
slurm_job_id, array_spec, requested_throttle, partition, account,
cpus_per_task, memory_per_task, time_limit, submitted_utc,
sacct_snapshot_path, expected_tasks, completed_tasks, failed_tasks,
peak_maxrss, aggregate_cpu_time, harvest_sha256, harvest_utc
```

The Speed option is verified only when Q1–Q3 have passed, the control audit has unlocked Q4, all expected outputs are harvested locally, and a clean-room recount reproduces the manifest totals. Availability of more than 32 CPUs is an execution opportunity, not evidence that the pipeline is scientifically valid.